Notes on the continuum sigma limit

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To start our analysis, consider a linear differential operator L acting on $\mathbb{R}^n$, and its Green's function $G(\mathbf{r}, \mathbf{r}')$, defined by

$$LG(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}'). \tag{1}$$

Introduce a variable τ, and define a heat kernel $K(\tau, \mathbf{r}, \mathbf{r}')$ by the equation

$$\frac{\partial}{\partial \tau} K(\tau, \mathbf{r}, \mathbf{r}') = LK(\tau, \mathbf{r}, \mathbf{r}'), \tag{2}$$

subject to the initial conditions $K(0, \mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}')$. If the operator L is dissipative in the sense that $\lim_{\tau \to \infty} K(\tau, \mathbf{r}, \mathbf{r}') = 0$, then we can readily prove the relation

$$G(\mathbf{r}, \mathbf{r}') = -\int_0^\infty K(\tau, \mathbf{r}, \mathbf{r}'). \tag{3}$$

To prove this relation, act with L on both sides, to find

$$LG(\mathbf{r}, \mathbf{r}') = -\int_0^\infty \frac{\partial}{\partial \tau} K(\tau, \mathbf{r}, \mathbf{r}') = -K(\tau, \mathbf{r}, \mathbf{r}')\Big|_0^\infty = \delta(\mathbf{r} - \mathbf{r}'). \tag{4}$$

Here we will consider a continuum approach to the multiple length scale splitting of the Coulomb interaction. The multiple length scale splitting of the potential is defined as follows

$$\frac{1}{4\pi r} = \sum_{i=0}^{n} v_i(r), \tag{5}$$

where v_i is defined as

$$v_i(r) = \frac{\text{erf}(r/\sigma_i) - \text{erf}(r/\sigma_{i+1})}{4\pi r}, \tag{6}$$

for some increasing sequence of positive numbers σ_i, and where $\sigma_0 = 0$ and $\sigma_{n+1} = \infty$, so that $v_0(r) = \frac{1 - \text{erf}(r/\sigma_1)}{r}$ and $v_n(r) = \frac{\text{erf}(r/\sigma_n)}{r}$. We will be interested in the continuum limit of this splitting, so that $n \to \infty, \sigma_n \to \infty$ and the spacing $d\sigma \equiv \sigma_i - \sigma_{i+1} \to 0$. In this limit, we can Taylor expand the error function around the value r/σ_i and keep only the first term,

$$\text{erf}\left(\frac{r}{\sigma + d\sigma}\right) \approx \text{erf}(r/\sigma) + d\sigma \frac{\partial}{\partial \sigma} \text{erf}\left(\frac{r}{\sigma}\right) = \text{erf}(r/\sigma) - \frac{2r d\sigma}{\sigma^2 \sqrt{\pi}} e^{-r^2/\sigma^2}. \tag{7}$$

Using this result, equation (6) becomes

$$v(r,\sigma) = \frac{1}{2\sigma^2\pi^{3/2}} e^{-r^2/\sigma^2}, \tag{8}$$

where we have replaced the discrete index i by the continuum index σ, and divided by $\mathrm{d}\sigma$ for the continuum limit. Using the change of variables $z = r/\sigma$, one can readily check the identity

$$\frac{1}{4\pi r} = \int_0^\infty v(r,\sigma) d\sigma, \tag{9}$$

which is the continuum limit of equation (5).

Physically, the function $v(r,\sigma)$ is a potential function in three dimensions, where r is the radius $r = \sqrt{x^2 + y^z + z^2}$. In fact, it is interesting to introduce a time like variable $4\tau = \sigma^2$, with which equation (8) becomes

$$v(\mathbf{r},\tau) = (4\pi\tau)^{-3/2} e^{-\mathbf{r}^2/4\tau}, \tag{10}$$

which again satisfies

$$\frac{1}{4\pi|\mathbf{r}|} = \int_0^\infty v(\mathbf{r},\tau) d\tau. \tag{11}$$

One readily recognizes equation (10) as the fundamental solution to the heat equation, or the heat kernel, so that

$$\frac{\partial}{\partial\tau} v(\mathbf{r},\tau) = \Delta v(\mathbf{r},\tau), \tag{12}$$

with the initial conditions $v(\mathbf{r},0) = \delta(\mathbf{r})$. With this result, it is natural to define a transform of a function via convolution with the function $v(\mathbf{r},t)$. Explicitly, for a function $f(\mathbf{r})$, the transform would be defined by

$$\tilde{f}(\mathbf{r},\tau) = \int f(\mathbf{r}') v(\mathbf{r}-\mathbf{r}',\tau) d^3\mathbf{r}', \tag{13}$$

which is equivalent to the generalized Weierstrass transform. Naturally, this transformed function would satisfy the heat equation

$$\frac{\partial}{\partial\tau} \tilde{f}(\mathbf{r},\tau) = \Delta \tilde{f}(\mathbf{r},\tau), \tag{14}$$

with initial conditions $\tilde{f}(\mathbf{r},0) = f(\mathbf{r})$. If $f(\mathbf{r})$ were a charge density $\rho(\mathbf{r})$, the coulomb potential would be given by the integral

$$V(\mathbf{r}) = \int \tilde{\rho}(\mathbf{r},\tau) d\tau, \tag{15}$$

which satisfies Poisson's equation

$$\Delta V(\mathbf{r}) = -\rho(\mathbf{r}), \tag{16}$$

recovering the original function from the transform.

This transform is useful in the study of quantum mechanical systems interacting under the Coulomb force. To start, write the molecular Hamiltonian in the form $\hat{H} = \hat{T} + \hat{V} + \hat{V}_{ee}$, where

$$\hat{T} = \int d^3r \, a^\dagger_\sigma(\mathbf{r}) \frac{\hat{p}^2}{2m} a_\sigma(\mathbf{r}), \quad \hat{V} = \int d^3r \, V(\mathbf{r}) \hat{\rho}(\mathbf{r}), \tag{17}$$

where $\hat{\rho}(\mathbf{r}) = a^\dagger_\sigma(\mathbf{r}) a_\sigma(\mathbf{r})$ is the density operator, and $\hat{V}_{ee}$ is the second quantized Coulomb potential

$$\hat{V}_{ee} = 2\pi e^2 \int d^3r \int d^3r' \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|} : \hat{\rho}(\mathbf{r}) \hat{\rho}(\mathbf{r}') :, \tag{18}$$

where : : denotes normal ordering. Here $V(\mathbf{r})$ includes the ionic potential, which will be treated as approximately fixed, and repeated indices are summed over.

To make contact with the previous analysis, define the Green's function operator

$$\hat{G} = -\int d^3r \int d^3r' \frac{1}{4\pi|\mathbf{r} - \mathbf{r}'|} : \hat{\rho}(\mathbf{r}) \hat{\rho}(\mathbf{r}') :, \tag{19}$$

so that the Coulomb operator can be written

$$\hat{V}_{ee} = 2\pi e^2 \hat{G}. \tag{20}$$

Also, we define a "heat kernel operator"

$$\hat{K}(\tau) = \int d^3r \int d^3r' K(\tau, \mathbf{r}, \mathbf{r}') : \hat{\rho}(\mathbf{r}) \hat{\rho}(\mathbf{r}') :, \tag{21}$$

which leads to an operatorial generalization of equation (11)

$$\hat{G} = -\int_0^\infty d\tau \hat{K}(\tau). \tag{22}$$

Here it is interesting to use the identity

$$K(\tau, \mathbf{r}, \mathbf{r}') = \int d^3\tilde{r} \, K(\tau/2, \mathbf{r}, \tilde{\mathbf{r}}) K(\tau/2, \tilde{\mathbf{r}}, \mathbf{r}'), \tag{23}$$

to rewrite these operators in terms in terms of "smoothed density operators". Explicitly, define the operator

$$\hat{\rho}(\tau, \mathbf{r}) = \int d^3r' \, K(\tau, \mathbf{r}, \mathbf{r}') \hat{\rho}(\mathbf{r}'), \tag{24}$$

with which equation (21) becomes

$$\hat{K}(\tau) = \int d^3r : \hat{\rho}(\tau/2, \mathbf{r})\hat{\rho}(\tau/2, \mathbf{r}) :, \tag{25}$$

which is formally local in the smooth charge density.

Equation (23) defines a generalized Weierstrass transform of the operator $\hat{\rho}(\mathbf{r})$. Formally, the generalized Weierstrass transform can be written as

$$W_\tau = e^{\tau\Delta}, \tag{26}$$

where Δ is the Laplacian operator. In the present context, it is natural to search for an operatorial generalization of this relation. To start, we note that the naïve Laplacian operator $\int d^3r\, a^\dagger_\sigma(\mathbf{r})\Delta a_\sigma(\mathbf{r})$ does not act as a Laplacian on the density operator $\hat{\rho}(\mathbf{r})$, only on the creation operator $a^\dagger_\sigma(\mathbf{r})$. To define a more relevant operator, we first define a gradient superoperator

$$\hat{\nabla} = \int d^3r\, [\, a^\dagger_\sigma(\mathbf{r})\nabla a_\sigma(\mathbf{r}),\, \cdot\,] \tag{27}$$

where the term superoperator is used because this acts on operators. With this operator, one can show

$$\hat{\nabla} \cdot \hat{\nabla}\hat{\rho}\,(\mathbf{r}) = \Delta\hat{\rho}(\mathbf{r}), \tag{28}$$

which defines a natural Laplacian superoperator. This leads to an operatorial generalization of the heat equation, but there are some properties that are left to be desired. One of the main difficulties is that the superoperator does not satisfy the Leibniz relation $\hat{\Delta}(\hat{A}\hat{B}) \neq \hat{\Delta}(\hat{A})\hat{B} + \hat{A}\hat{\Delta}(\hat{B})$, which tends to complicate calculations. Luckily, there is a related operator that can be defined and leads to this Laplacian operator as an "even subalgebra", in a sense to be explained below.

To this end, define the operator

$$\hat{\not\partial} = \sum_{i=1}^{3} \int d^3r\, a^\dagger_\sigma(\mathbf{r})\sigma_i \frac{\partial}{\partial r_i} a_\sigma(\mathbf{r}), \tag{29}$$

where here σ_i are the Pauli sigma matrices, and we have introduced Feynman slash notation, as this is a generalization of the usual three dimensional $\not\partial$ operator. We will want to prove the following identity, which will lead to useful simplifications in our analysis

$$\hat{\rho}(s^2, \mathbf{r}) = \mathrm{Tr}\left(e^{s\hat{\not\partial}}\hat{\rho}(\mathbf{r})e^{-s\hat{\not\partial}}\right), \tag{30}$$

which we argue is the appropriate operatorial generalization of equation (26). To prove this, we use an adaptation of the proof of the combinatorial lemma $\mathrm{Ad}_{e^X} = e^{\mathrm{ad}_X}$ used in the proof of the Baker-Campbell-Hausdorff formula [1]. First, define the superoperator $f(s)\hat{Y} = e^{s\hat{\not\partial}}\hat{Y}e^{-s\hat{\not\partial}}$,

and take the trace of the second derivative with respect to s of $f(s)\hat{Y}$, where $\hat{Y}$ is taken to be proportional to the identity matrix in spin space

$$\mathrm{Tr}\Big(\frac{\partial^2}{\partial s^2} f(s)\hat{Y}\Big) = \hat{\nabla}\cdot\hat{\nabla}\big(f(s)\hat{Y}\big). \tag{31}$$

This must be true of any such operator $\hat{Y}$, so $f(s)$ must be a solution of the differential equation

$$\frac{\partial^2}{\partial s^2} f(s) = \hat{\nabla}\cdot\hat{\nabla} f(s), \tag{32}$$

subject to the initial conditions $f(0) = 1$ and $f'(s)|_{s=0} = 0$. The unique solution of this differential equation is

$$f(s) = \cosh\left(s\hat{\nabla}\cdot\hat{\nabla}\right), \tag{33}$$

In order to understand this rewriting of the Coulomb potential in more depth, it is useful to consider how it acts on a wavefunction. To this end, we first define the Laplacian of $\hat{K}(\tau)$ as

$$\hat{\nabla} = \int d^3r\, [\, a^\dagger_\sigma(\mathbf{r})\nabla a_\sigma(\mathbf{r}),\, \cdot\,] \tag{34}$$

$$\hat{\Delta} = \hat{\nabla}\cdot\hat{\nabla}, \tag{35}$$

$$\Delta\hat{K}(\tau) = \int d^3r \int d^3r' K(\tau,\mathbf{r},\mathbf{r}') : \hat{\rho}(\mathbf{r})\Delta\hat{\rho}(\mathbf{r}') :, \tag{36}$$

which implies the generalized heat equation

$$\frac{\partial}{\partial\tau}\hat{K}(\tau) = \Delta\hat{K}(\tau), \tag{37}$$

with the solution

$$\hat{K}(\tau) = e^{\int d\tau \Delta\hat{K}(\tau)} \tag{38}$$

With this definition, a short calculation shows

$$\big[\hat{\Delta}, \hat{\rho}(\mathbf{r})\big] = \Delta\hat{\rho}(\mathbf{r}), \tag{39}$$

which implies the relation

$$\frac{\partial}{\partial\tau}\hat{K}(\tau,\mathbf{r},\mathbf{r}') = \big[\hat{\Delta}, \hat{K}(\tau,\mathbf{r},\mathbf{r}')\big], \tag{40}$$

We will then define the following Laplacian super-operator

$$\hat{\Delta}_{\mathbf{r}} = \lim_{\tau \to 0} \int d^3r \int d^3r' K(\tau, \mathbf{r}, \mathbf{r}') \hat{\nabla}_{\mathbf{r}} \cdot \hat{\nabla}_{\mathbf{r}'}. \tag{41}$$

With this formalism, one can verify the identity

$$\frac{\partial}{\partial \tau} \hat{K}(\tau, \mathbf{r}, \mathbf{r}') = \hat{\Delta}(\hat{K}(\tau, \mathbf{r}, \mathbf{r}')). \tag{42}$$

which is the operatorial version of the heat equation. The inverse Weierstrass operator W_τ^{-1} is given by $W_{-\tau}$. This operator imposes a Weirstrass transform under conjugation

$$\hat{W}_\tau a^\dagger(\mathbf{r}) \hat{W}_\tau^{-1} = \int d^3r' K(\tau/4, \mathbf{r}, \mathbf{r}') a^\dagger(\mathbf{r}'). \tag{43}$$

When acting on a multi-particle state, one readily sees that it acts on each state separately

$$\hat{W}_\tau a^\dagger(\mathbf{r}_1) a^\dagger(\mathbf{r}_2) \ldots = (\hat{W}_\tau a^\dagger(\mathbf{r}_1) \hat{W}_\tau^{-1})(\hat{W}_\tau a^\dagger(\mathbf{r}_2) \hat{W}_\tau^{-1}) \ldots, \tag{44}$$

and so acts as a smoothing operator for any multiparticle state.

This operator is useful for simplifying the Coulomb expression. Starting with equation (**??**), one can show

$$\hat{V}_{ee} = \frac{\pi e^2}{2} \int d\tau \int d^3r \, \hat{W}_\tau^{-1} \hat{\rho}(\mathbf{r}) \hat{\rho}(\mathbf{r}) \hat{W}_\tau. \tag{45}$$

To see why this is true, we consider the action of this operator on a multiparticle state. First, the operator smooths the wavefunction with a Gaussian through W_τ, then multiplies by the total squared smooth density, and then unsmooths the wavefunction with the inverse transform. So effectively it multiplies by the squared smooth charge density. From equation (23), we see that the original expression does the same thing. Because the two operators act linearly on superpositions of states, and we have argued that they should give the same result when acting on any multiparticle state, they should be equal. Although this derivation can be made rigorous, a more direct derivation using operator algebra is also presented in appendix A.

In these equations, the parameter τ determines an effective length scale of interaction. This is reminiscent of the renormalization group, and with this motivation in mind it is natural to define a "truncated" Coulomb operator by the integral

$$\hat{V}_{ee}(\tau) = \frac{\pi e^2}{2} \int_\tau^\infty d\tau' \int d^3r : \hat{\rho}(\tau', \mathbf{r}) \hat{\rho}(\tau', \mathbf{r}) : . \tag{46}$$

Clearly this potential reproduces the exact Coulomb operator at $\tau = 0$. The ionic potential operator $\hat{V}$ can also be truncated in the same manner, as this potential also arises from a Coulomb interaction. We can therefore define a truncated Hamiltonian by

$$\hat{H}(\tau) = \hat{T} + \hat{V}(\tau) + \hat{V}_{ee}(\tau), \tag{47}$$

which again reproduces the full Hamiltonian at $\tau = 0$. In the opposite limit $\tau \to \infty$, the truncated interaction vanishes entirely, which implies

$$\lim_{\tau\to\infty} \hat{H}(\tau) = \hat{T}, \tag{48}$$

describing the free electron Hamiltonian.

$$\hat{W}_\tau = \exp\Big(\tau \int d^3r\, a^\dagger_\sigma(\mathbf{r})\Delta a_\sigma(\mathbf{r})\Big), \tag{49}$$

To understand the transition between the limits discussed above, we will first consider a solution to the time dependent Schrodinger equation as a function of τ

$$i\hbar\frac{\partial}{\partial t}\left|\psi(\tau,t)\right\rangle = \hat{H}(s)\left|\psi(\tau,t)\right\rangle, \tag{50}$$

with the intial condition $|\psi(tau,0)\rangle = |\psi_0\rangle$. In order to be well defined, this

A Coulomb operator to Weierstrass transform

$$[\hat{K}(\tau,\mathbf{r},\mathbf{r}'),\hat{\rho}(\tilde{\mathbf{r}})] = [\hat{K}(\tau,\mathbf{r},\mathbf{r}'),a^\dagger_{\tilde{\sigma}}(\tilde{\mathbf{r}})]a_{\tilde{\sigma}}(\tilde{\mathbf{r}}) + a^\dagger_{\tilde{\sigma}}(\tilde{\mathbf{r}})[\hat{K}(\tau,\mathbf{r},\mathbf{r}'),a_{\tilde{\sigma}}(\tilde{\mathbf{r}})]$$
$$[\hat{K}(\tau,\mathbf{r},\mathbf{r}'),a^\dagger_\sigma(\tilde{\mathbf{r}})] = \int d^3r d^3r' K(\tau,\mathbf{r},\mathbf{r}')[:\hat{\rho}(\mathbf{r})\hat{\rho}(\mathbf{r}'):,a^\dagger_\sigma(\tilde{\mathbf{r}})]$$

$$\int d^3r d^3r'[:\hat{\rho}(\mathbf{r})\hat{\rho}(\mathbf{r}'):,a^\dagger_{\tilde{\sigma}}(\tilde{\mathbf{r}})] = \int d^3r d^3r'[a^\dagger_\sigma(\mathbf{r})a^\dagger_{\sigma'}(\mathbf{r}')a_{\sigma'}(\mathbf{r}')a_\sigma(\mathbf{r}),a^\dagger_{\tilde{\sigma}}(\tilde{\mathbf{r}})] =$$
$$\int d^3r d^3r'\Big(a^\dagger_\sigma(\mathbf{r})a^\dagger_{\sigma'}(\mathbf{r}')a_{\sigma'}(\mathbf{r}')\{a_\sigma(\mathbf{r}),a^\dagger_{\tilde{\sigma}}(\tilde{\mathbf{r}})\} - a^\dagger_\sigma(\mathbf{r})a^\dagger_{\sigma'}(\mathbf{r}')\{a_{\sigma'}(\mathbf{r}'),a^\dagger_{\tilde{\sigma}}(\tilde{\mathbf{r}})\}a_\sigma(\mathbf{r})\Big) =$$
$$2\int d^3r\, a^\dagger_{\tilde{\sigma}}(\tilde{\mathbf{r}})\hat{\rho}(\mathbf{r})$$

$$\int d^3r d^3r'[:\hat{\rho}(\mathbf{r})\hat{\rho}(\mathbf{r}'):,a_{\tilde{\sigma}}(\tilde{\mathbf{r}})] = \int d^3r d^3r'[a^\dagger_\sigma(\mathbf{r})a^\dagger_{\sigma'}(\mathbf{r}')a_{\sigma'}(\mathbf{r}')a_\sigma(\mathbf{r}),a_{\tilde{\sigma}}(\tilde{\mathbf{r}})] =$$
$$\int d^3r d^3r'\Big(-\{a_{\tilde{\sigma}}(\tilde{\mathbf{r}}),a^\dagger_\sigma(\mathbf{r})\}a^\dagger_{\sigma'}(\mathbf{r}')a_{\sigma'}(\mathbf{r}')a_\sigma(\mathbf{r}) + a^\dagger_\sigma(\mathbf{r})\{a_{\tilde{\sigma}}(\tilde{\mathbf{r}}),a^\dagger_{\sigma'}(\mathbf{r}')\}a_{\sigma'}(\mathbf{r}')a_\sigma(\mathbf{r})\Big) =$$
$$-2\int d^3r\hat{\rho}(\mathbf{r})a_{\tilde{\sigma}}(\tilde{\mathbf{r}})$$

$$\int d^3r d^3r' [: \hat{\rho}(\mathbf{r})\hat{\rho}(\mathbf{r}') :, : \hat{\rho}(\tilde{\mathbf{r}})\hat{\rho}(\tilde{\mathbf{r}}') :] = 0 \tag{51}$$

$$\int d^3r d^3r' K(\tau, \mathbf{r}, \mathbf{r}') K(\tau, \tilde{\mathbf{r}}, \tilde{\mathbf{r}}') : \hat{\rho}(\mathbf{r})\hat{\rho}(\mathbf{r}') :: \hat{\rho}(\tilde{\mathbf{r}})\hat{\rho}(\tilde{\mathbf{r}}') := \tag{52}$$

In this appendix, we prove equation (45). To start, we note the commutation relations

$$\Big[\int d^3r' \, a^\dagger_{\sigma'}(\mathbf{r}') \Delta a_{\sigma'}(\mathbf{r}'), a^\dagger_\sigma(\mathbf{r}) \Big] = \Delta a^\dagger_\sigma(\mathbf{r}), \tag{53}$$

$$\Big[\int d^3r' \, a^\dagger_{\sigma'}(\mathbf{r}') \Delta a_{\sigma'}(\mathbf{r}'), a_\sigma(\mathbf{r}) \Big] = -\Delta a_\sigma(\mathbf{r}), \tag{54}$$

where we have assumed that all boundary terms vanish when integrating by parts. Using the formula [1]

$$e^X Y e^{-X} = Y + [X, Y] + \frac{1}{2!}[X, [X, Y]] + \frac{1}{3!}[X, [X, [X, Y]]] + \ldots, \tag{55}$$

one can readily prove the relation

$$\partial_\tau \Big(\hat{W}_\tau a^\dagger_\sigma(\mathbf{r}) \hat{W}^{-1}_\tau \Big) = 4\Delta \Big(\hat{W}_\tau a^\dagger_\sigma(\mathbf{r}) \hat{W}^{-1}_\tau \Big), \tag{56}$$

which is solved by the convolution

$$\hat{W}_\tau a^\dagger_\sigma(\mathbf{r}) \hat{W}^{-1}_\tau = \int d^3r' K(\tau/4, \mathbf{r}, \mathbf{r}') a^\dagger_\sigma(\mathbf{r}'), \tag{57}$$

which justifies calling this a smoothing operator. To prove equation (45), we first use equations (53) and (54) to find

$$\int d^3r \Big[\int d^3r' \, a^\dagger_{\tilde{\sigma}}(\mathbf{r}') \Delta a_{\tilde{\sigma}}(\mathbf{r}'), a^\dagger_\sigma(\mathbf{r}) a^\dagger_{\sigma'}(\mathbf{r}) a_{\sigma'}(\mathbf{r}) a_\sigma(\mathbf{r}) \Big] = \tag{58}$$

$$\int d^3r \Big((\Delta a^\dagger_\sigma(\mathbf{r})) a^\dagger_{\sigma'}(\mathbf{r}) a_{\sigma'}(\mathbf{r}) a_\sigma(\mathbf{r}) + a^\dagger_\sigma(\mathbf{r}) (\Delta a^\dagger_{\sigma'}(\mathbf{r})) a_{\sigma'}(\mathbf{r}) a_\sigma(\mathbf{r}) \tag{59}$$

$$- a^\dagger_\sigma(\mathbf{r}) a^\dagger_{\sigma'}(\mathbf{r}) (\Delta a_{\sigma'}(\mathbf{r})) a_\sigma(\mathbf{r}) - a^\dagger_\sigma(\mathbf{r}) a^\dagger_{\sigma'}(\mathbf{r}) a_{\sigma'}(\mathbf{r}) (\Delta a_\sigma(\mathbf{r})) \Big) \tag{60}$$

References

[1] B. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*. Graduate Texts in Mathematics. Springer, 2003.